The SX Phoenicis Distance Project

A Field Guide, Physical Primer, and Methods Reference

Companion to the period–luminosity distance measurement of CY Aquarii

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This document is meant to be read front to back — it begins with the big picture (what variable stars are and why they matter), works through the physics of stellar pulsation from intuition to detail, tells the history of how a flickering star became a cosmic yardstick, and closes with a thorough account of the measurement methods used in the project. Each section is layered: skim the opening paragraphs for the concept, or read on for the depth. A companion notebook, *equations_reference.ipynb*, carries every key formula with worked numbers and runnable code.

The project in one sentence. Using freely available archival photometry (AAVSO) and satellite parallaxes (Gaia), we measure the distance to the pulsating star CY Aquarii by exploiting the fact that its pulsation period reveals its true brightness — the same principle that underlies humanity's measurement of the scale of the universe.

Part I · Introduction

Variable stars, the pulsators, and the law that turns a flicker of light into a distance.

A field guide to variable stars

A variable star is any star whose brightness changes over time in a way we can measure. Astronomers sort them into two great families by the *cause* of the change. **Extrinsic** variables do not actually change their light output; something external modulates what we see — an orbiting companion eclipsing the star, or starspots rotating in and out of view. **Intrinsic** variables genuinely change their own luminosity, and these split further into eruptive stars (flares, outbursts), cataclysmic stars (novae and supernovae), and — the family at the heart of this project — **pulsating** stars, which rhythmically expand and contract at their own natural frequencies.

Pulsation is not a quirk of a few odd stars; it is a natural phase many stars pass through when their internal structure lets a particular instability take hold. Part III develops the physics in detail: the oscillation is resonant (the star has a natural frequency set by its own size and density), self-excited (an internal heat engine resupplies it each cycle), and diagnostic. That last property is the one that matters here — the pulsation period encodes the star's true luminosity, which is what makes pulsators measuring sticks for the cosmos.

What this project does

CY Aquarii is a pulsating star of the SX Phoenicis class — a fast, faint, metal-poor pulsator (Part II explains exactly what that means). The exercise is a **validate-first** investigation: prove an entire distance-measurement method on existing public data, checking every step against an independent answer key, before ever pointing a telescope. The pipeline runs in stages, each detailed in Part V:

- **Measure the light curve** — many brightness measurements over time.
- **Recover the period** — find the repeating pulsation cycle.
- **Fold and average** — compute the correct mean brightness.
- **Period → luminosity** — apply the period–luminosity relation.
- **Luminosity → distance** — via the distance modulus.
- **Check against Gaia** — the satellite's independent parallax distance.

The result: CY Aqr's distance, measured from a backyard-accessible method, agrees with the Gaia satellite's parallax to within the uncertainties — and along the way, the project independently re-derives the published period–luminosity relation from eight stars. That is the whole arc: a flickering point of light, turned into a distance of roughly 1,350 light-years, and a verification that the method works.

Part II · The Variable Star Zoo

A tour of the variable families — each in real depth — and a close study of the two classes this project lives in.

The intrinsic variables, in detail

Because pulsation is only one of three ways a star can genuinely change its own light, it helps to see the whole intrinsic family first. The distinction between them is *what physically changes*.

Pulsating variables. The star rhythmically expands and contracts. As its radius and surface temperature swing, so does its brightness. Some pulsate *radially* (the whole star oscillating symmetrically), others *non-radially* (patterns of rising and falling patches, like tides interfering on a sphere). Many pulsate in several modes at once. The period is locked to the star's physical structure — which is what makes period–luminosity relations, and this project, possible.

Eruptive variables. Light changes because of violent activity in the star's atmosphere or envelope — magnetic flares, unstable winds, shell ejections — not because the star as a whole restructures. Flare stars, T Tauri stars, and luminous blue variables live here; their changes are often irregular, tracking surface chaos rather than a clock.

Cataclysmic variables. The explosive ones, almost always a binary in which a white dwarf pulls material off a companion. Accreted gas can ignite in a thermonuclear flash (a **nova**); disk instabilities give recurrent dwarf-nova outbursts; and at the extreme sits the **supernova**. Type Ia supernovae are standard candles in their own right, anchoring the far end of the same distance ladder the pulsators serve nearer home.

A field guide to the pulsating classes

The pulsators of the **instability strip** form a family sequence — same engine, wildly different stars. To place them mentally, picture the Hertzsprung–Russell diagram: luminosity on the vertical axis (brighter upward), surface temperature on the horizontal (hotter to the *left*, by historical convention). Most stars lie along the diagonal **main sequence**, running from hot luminous stars at upper-left to cool faint ones at lower-right; evolved **red giants** branch off to the upper-right, and spent **white dwarfs** sit isolated at lower-left. The **instability strip** cuts across all of this as a narrow, nearly vertical band at around 6,000–8,000 K — a corridor of temperature, spanning an enormous range of luminosity. The pulsator classes are stations along that corridor, from luminous supergiants at the top down to faint dwarfs where the strip crosses the main sequence itself. Here is each in enough depth to recognise it. One class below sits *outside* the strip entirely, on a different engine, and is flagged as such when we reach it.

Classical Cepheids. Luminous yellow supergiants — evolved stars of 4–20 solar masses that have left the main sequence and are looping through the strip as they burn helium.

Periods run from about 1 to over 100 days; luminosities reach tens of thousands of Suns, so they are visible in other galaxies — which is precisely why Leavitt's Law (Part IV) made them the aristocrats of the distance ladder. Their light curves are smooth and asymmetric, and their period–luminosity relation is the tightest of any class. A Cepheid is what a massive star briefly becomes; the phase lasts mere millions of years, so Cepheids are always young, disk-population objects.

Type II Cepheids. Superficially similar periods (1–50 days) but a different animal: old, metal-poor, *low-mass* stars (under a solar mass) in late evolutionary stages, found in the halo and globular clusters. They are roughly 1.5–2 magnitudes fainter than classical Cepheids at the same period — a historical trap: early distance work unknowingly mixed the two classes, miscalibrating the extragalactic scale until Walter Baade separated the populations in the 1950s. Their existence is the classic warning that population matters in PL relations — the same warning that runs through this project's mixing of SX Phe and δ Scuti.

RR Lyrae. The workhorses of old stellar systems: horizontal-branch stars — old, low-mass stars that have ignited helium in their cores — passing through the strip at nearly uniform luminosity (about 40–50 Suns). Periods cluster tightly between 0.2 and 1 day. Because their absolute magnitude is nearly constant ($M_V \approx +0.6$), they act almost as standard candles *without* needing a period measurement, and they swarm in globular clusters — making them the primary rulers for cluster distances and ages, and tracers of the galaxy's ancient halo.

δ Scuti. Down near the base of the strip, where it crosses the main sequence: stars of 1.5–2.5 solar masses, spectral types A to early F, either still on the main sequence or just leaving it. Periods of 0.02–0.3 days (half an hour to seven hours); most pulsate in a froth of low-amplitude non-radial modes simultaneously (a gift to asteroseismology), but a subset — the **high-amplitude δ Scuti (HADS)** — pulsate in one or two dominant radial modes with swings above 0.3 mag, giving clean sawtooth light curves. These are young-to-middle-aged, metal-rich Population I disk stars.

SX Phoenicis. The metal-poor floor of the strip and this project's home class — treated in its own section below.

Mira variables — the deliberate outsider. Huge, cool red giants (spectral type M) in the last stages of life, pulsating with periods of 100–1000 days and visual amplitudes that can exceed 8 magnitudes — the largest of any class. Crucially, they sit far to the cool side of the H–R diagram, well *outside* the instability strip — and that placement is physical, not incidental: Miras are **not** driven by the same helium-ionization valve. Their pulsation is powered by hydrogen ionization interacting with the deep convection of a bloated envelope — a related but distinct engine. They are the standing reminder that not all pulsators are κ-mechanism pulsators.

Which variables does the κ-mechanism actually drive? A point worth making crisply, because it is easy to over-generalise. The classic helium-ionization κ-mechanism of Part III drives the **instability-strip family**: classical and Type II Cepheids, RR Lyrae, δ Scuti, and SX Phe. It does *not* drive everything that varies. Miras pulsate via hydrogen ionization coupled to convection; β Cephei and other hot B-star pulsators run on a κ-mechanism too, but powered by an iron-opacity bump at much higher temperature; white-dwarf pulsators are driven by partial ionization in their thin surface layers; the Sun's five-minute oscillations are excited stochastically by convective noise; and eruptive and cataclysmic variables involve no pulsation at all. So: one strip, one valve — but many other ways for a star to vary.

δ Scuti and SX Phoenicis: the same engine, two populations

Now the close-up on the two classes this project draws on. Physically, a high-amplitude δ Scuti and an SX Phe star are nearly the same object: 1–2.5 solar masses, sitting where the instability strip crosses the main sequence, pulsating in one or two radial modes with periods of one to a few hours. Both produce the same sawtooth light curve. If you only had a light curve, you often could not tell them apart.

On that H–R picture, the two classes occupy overlapping territory at the base of the strip — δ Scuti slightly more luminous on average, SX Phe reaching a little fainter and hotter. Their positions overlap because the difference between them is not where they sit but what they are made of.

The one difference that matters: metallicity, hence population. δ Scuti stars are metal-rich Population I — young disk stars with roughly the Sun's complement of heavy elements. SX Phoenicis stars are metal-poor Population II — ancient halo and globular-cluster stars with iron abundances 10 to 30 times below solar ([Fe/H] ≈ −1.5 to −1.0). They are the same pulsator drawn from two different eras of the galaxy's history.

Why population matters for distances. Metallicity subtly changes stellar structure and opacity, shifting the luminosity at a given period. So the two classes share nearly the same period–luminosity *slope* (set by pulsation physics, Part III) but slightly different *zero-points* (set by chemistry). The strict relation is really $M_V = a + b \cdot \log P + c \cdot [Fe/H]$. Mixing the populations in one fit adds scatter — which is exactly why the reference calibration (Cohen & Sarajedini 2012) used pure SX Phe, and why this project's own 8-star fit, which mixes both, shows extra spread. The Type II Cepheid story above is the same lesson at higher luminosity.

The blue-straggler twist. Here is what makes SX Phe genuinely strange. In an old globular cluster, every star formed at once, so the massive blue ones died long ago — yet **blue stragglers** persist, apparently rejuvenated. The leading explanation: two old, low-mass stars *merged* (or one stripped mass from a companion), building a single more-massive star that rejoined the blue main sequence. Many SX Phe are exactly these objects. A pulsating SX Phe is often a stellar merger caught in the act of ringing — simultaneously a probe of pulsation physics and of the collisional history of dense clusters. Masses run about 1.0–1.3 solar; the material is ancient even when the star looks young.

CY Aquarii in this context. Field SX Phe — bright, isolated ones outside clusters — are rare: only about fourteen are known, because the class lives overwhelmingly in globular clusters and dwarf galaxies where crowding defeats small-telescope photometry. CY Aqr is one of those rare bright field examples, and among the shortest-period members known (0.061 d, about 88 minutes). Brightness, isolation, and a fast clock: a single night captures several complete cycles, which is precisely what makes it the ideal first target.

Part III · The Physics of Pulsation

The resonance, the engine that sustains it, and why the period reveals the luminosity.

Level 1 — the shape of the problem

A pulsating star oscillates: its outer layers move rhythmically outward and inward, and its brightness rises and falls with them. Three properties of that oscillation matter, and each does real work later.

It is resonant. The star is not being driven at some externally imposed rhythm. It oscillates at its own *natural frequency* — a standing wave in a spherical cavity, with a frequency fixed by the physical properties of the cavity itself. Change the star's size or density and the frequency changes with it, in a way we can calculate exactly (Level 2).

It is self-excited. An oscillation in ordinary gas damps out; friction and radiative losses drain it within a few cycles. Pulsating stars have nonetheless been oscillating steadily for millions of years, which means the star contains an internal engine that resupplies the oscillation's energy on every cycle, exactly in phase. Identifying that engine — the κ-mechanism — is the business of Level 3.

It is diagnostic. Because the frequency is set by the star's bulk properties, measuring the period tells us something about the star's interior that we could not otherwise obtain. This is what elevates pulsation from a curiosity to a tool.

Those three facts assemble into the claim that drives this entire project: **the pulsation period determines the star's size, and its size determines its luminosity.** Measure how fast a star pulsates, and you have measured how bright it truly is — regardless of how bright it happens to appear from Earth. Compare true brightness to apparent brightness and the distance falls out. The remaining levels establish each link in that chain as physics rather than assertion: period → density (Level 2), the engine that sustains it (Level 3), why only certain stars qualify (Level 4), and density → size → luminosity (Level 5).

Level 2 — the pulsation clock: why period reads out density

Why does a larger star pulsate more slowly? Because the pulsation is, physically, a **sound wave**: a pressure disturbance travelling through the star's gas at the local speed of sound, reflecting at the surface and centre, setting up a standing wave. The period is essentially the round-trip time — the *sound-crossing time* of the star. A larger, more distended star takes longer to cross; a denser star has faster sound and a shorter crossing.

To see where the famous scaling comes from, reason it out rather than accept it. The sound-crossing time is roughly the radius divided by the sound speed, $t \sim R/c_s$. The sound speed in a self-gravitating ball is set by how strongly gravity squeezes the gas: dimensional analysis (or hydrostatic equilibrium) gives $c_s \sim \sqrt{GM/R}$. Put these together:

$$P \sim R / c_s \sim R / \sqrt{GM/R} = \sqrt{R^3/GM} \sim 1/\sqrt{G\bar{\rho}}$$

since mean density is $\bar{\rho} = M / (4\pi R^3/3)$. The mass and radius collapse into a single quantity — density — and the result is the **period–mean-density relation**, usually written with a dimensionless "pulsation constant" Q that absorbs the details of stellar structure and the mode:

$$P \sqrt{\bar{\rho}/\bar{\rho}_{\odot}} = Q \quad \Rightarrow \quad P \propto \bar{\rho}^{-1/2}$$

Q is nearly the same for all stars pulsating in a given mode (about 0.033 days for the fundamental radial mode), which is what makes the relation so powerful: **the period is a direct readout of mean density**, one of the most useful scaling laws in stellar physics. For CY Aqr, P = 0.061 d gives a mean density of roughly a third of the Sun's — a slightly distended star of about 1–2 solar radii, exactly as expected for its class. Denser stars pulsate faster; the period is, in effect, a densitometer that works across interstellar distances.

Level 3 — the engine: what sustains the oscillation

An oscillation in ordinary stellar gas damps out: radiative losses and turbulent friction drain its energy within a handful of cycles. Yet pulsating stars oscillate steadily for millions of years. Something must resupply the oscillation on every cycle, faster than damping drains it. That engine is the **κ-mechanism** — κ (kappa) being the standard symbol for *opacity*, the measure of how effectively stellar material blocks radiation. (In older literature the same idea appears as the *Eddington valve*, after Arthur Eddington, who first proposed that a stellar layer could act as a heat valve. The term is now largely obsolete, but is worth recognising if you meet it.)

The engine lives in a special layer. In *most* of a star, compressing the gas heats it, ionises it further, and makes it *more transparent* — so a compressed layer promptly leaks its heat, and any oscillation damps away. But in a **partial ionization zone** — a layer where a particular species is caught mid-way through losing an electron — the energy of compression goes into *finishing the ionization* rather than raising the temperature, and the gas responds by becoming *more opaque* when squeezed. That inverted response — opacity *rising* under compression — is the whole trick.

Which layer, precisely. Astronomers label ionization states with Roman numerals: **He I** is neutral helium, **He II** is singly-ionized (He^{+}), and **He III** is doubly-ionized (He^{++}). A star of spectral type A, F, or G has neutral helium in its photosphere. Descend inward and you cross two ionization boundaries in turn: at roughly **25,000–30,000 K** helium loses its first electron (He I → He II), and deeper still, at about **35,000–50,000 K**, it loses the second (He II → He III). It is this *second* transition — conventionally called the He II ionization zone, since it is where He II is being consumed — that drives the pulsation in instability-strip stars. Compression there supplies exactly the energy needed to strip that last electron, so the energy goes into ionization rather than into raising the temperature, and the resulting free electrons drive the opacity up. On expansion, He III recombines with free electrons back to He II, the opacity falls, and the trapped heat escapes outward.

Follow the cycle. The layer is squeezed as the star contracts — and being a partial-ionization zone, it turns opaque (panel 1). Opaque, it dams up the radiation streaming from the core; heat and pressure build behind it (panel 2). The trapped pressure shoves the layer outward, past its equilibrium point (panel 3). Expanded and decompressed, the zone turns transparent, the dammed heat floods out, pressure support collapses (panel 4) — and the layer falls back inward, overshoots, recompresses, and the cycle begins again.

Notice the thermodynamic shape of this: the layer **absorbs heat when compressed and releases it when expanded**. That is precisely the timing a heat engine needs to do net positive work per cycle — it is a piston engine in spirit, running on the star's own outflowing radiation, with the ionization zone as the valve. Each cycle, the valve delivers a fresh kick to the oscillation, sustaining it indefinitely against damping.

Scope check: which variables does this engine drive? The κ-mechanism is a *family* of drivers, not a single case: it can operate wherever opacity rises with compression, which happens in partial ionization zones of hydrogen or helium, around negative hydrogen ions (H^{-}), and at metal-opacity peaks. Which zone does the driving determines which stars pulsate.

The **helium** version described above — specifically the second ionization, He II → He III — powers the **instability-strip family**: classical and Type II Cepheids, RR Lyrae, δ Scuti, and SX Phe, including CY Aqr. **Hydrogen** ionization instead drives Mira variables (coupled to their deep convective envelopes), the rapidly oscillating Ap stars (roAp), and the ZZ Ceti pulsating white dwarfs. And in the hot **β Cephei** stars the driver is neither: it is a peak in *iron*-group opacity at a depth where the temperature reaches roughly 200,000 K, known as the **Z bump** (Z being the astronomical symbol for elements heavier than hydrogen and helium). Meanwhile the Sun's five-minute oscillations are not κ-driven at all — they are excited stochastically by convective turbulence — and the eruptive and cataclysmic variables of Part II involve no pulsation whatsoever.

Level 4 — the instability strip: why only some stars pulsate

If partial ionization zones exist in essentially every star — and they do — why doesn't every star pulsate? Because the valve only has leverage when it sits at the **right depth**, and its depth is set by the star's surface temperature. In a star that is too hot, the He II zone lies high in the atmosphere with too little mass above it to push against: the engine sputters without a load. In a star too cool, the outer layers convect — energy is carried by rising gas parcels rather than by radiation, so there is no radiative flow for the valve to throttle: the engine has no working fluid. Only in a narrow window of surface temperature (roughly 6,000–8,000 K at the strip's base) does the zone sit deep enough to drive and shallow enough to matter.

On the H–R diagram that temperature window is a narrow, nearly vertical band — the **instability strip**. A star pulsates when, and only when, its evolution carries it into the strip; classical Cepheids loop through it as evolved supergiants, RR Lyrae cross it on the horizontal branch, and δ Scuti / SX Phe stars sit where the strip crosses the main sequence itself. Two consequences matter for what follows: pulsation is a *phase*, not a permanent property; and because the strip is narrow, **every pulsator in it has nearly the same surface temperature** — the fact the next level turns into a distance tool.

Pulsation is a phase, not a property

That first consequence deserves unpacking, because it overturns a natural assumption. It is tempting to think of "variable star" as a category of object, like "red dwarf" or "white dwarf" — a kind of star you either are or are not. It is closer to a stage of life. A star is variable *while it happens to be in the strip*, the way a person is a teenager while they happen to be thirteen to nineteen.

Not every star gets the chance. A star's path across the H–R diagram is dictated almost entirely by its mass, with metallicity and mass loss as secondary influences. Some paths cross the instability strip; many never approach it. Low-mass red dwarfs spend hundreds of billions of years far to the cool side and never come near. The Sun sits just cool of the strip now and will pass below it as a red giant — it will never be a Cepheid or an RR Lyrae. Massive stars do cross, becoming classical Cepheids, but massive stars are rare. And whether an old low-mass star becomes an RR Lyrae depends on exactly how much envelope it shed as a red giant, which decides where on the horizontal branch it lands — two stars of identical initial mass can end up on opposite sides of the strip. Strip membership is a lottery decided by mass, chemistry, and evolutionary accident, and most stars never win it.

And those that do, leave. Pulsation is usually transient, sometimes startlingly so. Classical Cepheids are caught mid-"blue loop" — an evolutionary excursion during core helium burning that swings the star hot, then cool, then hot again. Each transit of the strip lasts perhaps a few thousand to a few hundred thousand years, against a stellar lifetime of tens of millions: a Cepheid typically crosses one to three times and then departs for good. **Polaris** is a case in progress — its pulsation amplitude declined markedly over the twentieth century, plausibly because it is evolving toward the strip's edge. RR Lyrae stars occupy the horizontal branch for roughly a hundred million years but can drift out of the unstable region as they evolve along it. δ Scuti and SX Phe stars, sitting where the strip crosses the main sequence, get the longest run — potentially hundreds of millions of years — but they too eventually exit as their temperature changes.

Why this matters for what we observe. If pulsation were permanent, the instability strip would be jammed with every star that ever entered it. Instead the strip holds a *steady-state* population: stars entering and leaving continuously, with the number present at any moment set by how long the phase lasts. Astronomers actually run this backwards — counting pulsators in a cluster of known age constrains how quickly stars cross the strip.

And this puts CY Aqr's history in a strange light. As a blue straggler (Part II), it did not reach the strip by ordinary evolution at all: two ancient, low-mass stars merged, producing a single more massive object that rejoined the main sequence — and happened to land inside the instability corridor. CY Aqr is pulsating tonight because of a stellar collision in the deep past. Its variability is not a scheduled milestone but an accident of dynamics, and one that will end when the star evolves out of the strip.

Level 5 — closing the loop: why period gives luminosity

Assemble the pieces. Level 2 established that the period reads out the star's mean density: longer period, lower density. At roughly similar masses (these stars span a narrow mass range), lower mean density can only mean one thing geometrically — a **larger radius**. So the period measures the size of the star.

The final link is the one law that converts size into light. Before writing it, build it: a star radiates from its surface, so its total output is (surface area) × (power emitted per unit area). The area of a sphere is $4\pi R^2$. And a hot surface radiates per unit area according to the Stefan–Boltzmann law, σT^4 — the same fourth-power-of-temperature law that makes a stove element glow furiously with a modest temperature rise. Multiply the two:

$$L = 4\pi R^2 \cdot \sigma T^4$$

Both radius and temperature matter — but Level 4 just told us that for instability-strip stars, **temperature is nearly fixed** (the strip is narrow). With T pinned, luminosity is governed by R^2 alone: bigger star, more light, full stop. The chain closes:

longer period → lower mean density (Level 2) → larger radius → at the strip's fixed temperature (Level 4), **higher luminosity** (Level 5).

Measure the period; infer the density; infer the radius; infer the luminosity. This is why a period–luminosity relation exists: it is the visible summary of a physical chain, not an empirical accident. Its negative slope in magnitudes ($M_V = a + b \cdot \log P$ with $b < 0$) is just the statement "longer period → brighter," since brighter means a more negative magnitude.

The relation is not perfectly tight, because each link has slack: mass varies somewhat from star to star; the strip has finite width, so temperature is not *exactly* fixed; and metallicity shifts the structure (Part II). Together these produce the **intrinsic scatter** — about 0.2 magnitudes — that no amount of careful photometry on a single star can beat. It is the precision floor of the whole method (Part V), and CY Aqr itself sits about 0.24 mag on the faint side of the mean relation: not an error, just an individual.

Reading the light curve as physics

One more payoff before the methods. The *shape* of the light curve encodes the engine directly. SX Phe and HADS stars show a characteristic **sawtooth** (Figure 1): a fast rise to maximum followed by a slow decline. CY Aqr climbs from minimum to maximum in about 20 minutes and takes nearly 70 to fall back.

Why asymmetric rather than a smooth sine? Because the star's *brightness* variation and its *radial* pulsation are not in step — there is a **phase shift** between them, and the size of that shift depends on how far the driving He II zone lies beneath the visible surface. The driving happens at depth; the light we see comes from the photosphere; and the energy released when the opacity valve opens takes time to travel outward through the intervening material. The resulting lag between the mechanical cycle and the luminosity cycle skews the light curve, producing the steep rise and gradual fall seen in most Cepheids and in the high-amplitude pulsators here. The asymmetry is therefore not incidental — it is a direct, observable consequence of the driving zone sitting at a particular depth below the surface.

Figure 1 — The sawtooth light curve in two colour bands. The star is hotter at maximum and cooler at minimum, so the B and V curves converge at peak and separate in the trough: the colour "breathes" with the temperature swing. The B amplitude is also larger, because hotter/cooler swings move more of the star's light in and out of the blue.

And because the star is genuinely **hotter at maximum, cooler at minimum** — the temperature swing of the pulsation itself — its *colour* changes through the cycle. Plot two bands together (Figure 1) and the gap between them widens and narrows each cycle: bluer at peak, redder in the trough, the instability-strip physics playing out every 88 minutes in real time. The blue band also swings with larger amplitude, since temperature changes shift the blue end of the spectrum hardest. This colour variation is small but real, and it is one reason precise photometry must handle the star's changing colour carefully (Part V, on transformation) — the star you are calibrating is not even the same colour from one phase to the next.

Part IV · Leavitt's Law: A History

How a flicker of light became a cosmic yardstick.

The discovery

The idea that a pulsator's period reveals its luminosity — and therefore its distance — came from **Henrietta Swan Leavitt** at the Harvard College Observatory in the early 1900s. Leavitt was one of the observatory's "computers," women hired to measure stars on glass photographic plates. Her assignment was to catalogue variable stars, and she focused on the **Small Magellanic Cloud (SMC)**, a companion galaxy to the Milky Way, using plates from Harvard's southern station in Arequipa, Peru.

Her decisive insight rested on simple geometry. All the stars in the SMC are, to good approximation, at the **same distance** from us — the cloud is far away and compact, like a swarm of fireflies seen from across a field. So any difference in their *apparent* brightness reflects a real difference in *intrinsic* luminosity, not distance. Cataloguing 1,777 variables in a 1908 paper and sharpening it with 25 Cepheids in her landmark 1912 paper, Leavitt noticed that the brighter variables took longer to cycle. In her own words, there was **a simple relation between the brightness of the variables and their periods**. Longer period, brighter star. Plotted against the logarithm of the period, the points fall along a line — the Period–Luminosity relation, now called **Leavitt's Law**.

The missing zero-point, and the ladder it built

Leavitt could rank the Cepheids by luminosity but could not put an **absolute** number on any of them, because the distance to the SMC itself was unknown. Her relation had the right *slope* but no **zero-point**. Within a few years **Ejnar Hertzsprung** and **Harlow Shapley** supplied it, calibrating the scale using nearer Cepheids in our own galaxy whose distances could be estimated by other means. Once anchored, the relation became a **standard candle**: measure a Cepheid's period, read off its true luminosity, compare to its apparent brightness, and the distance falls out of the inverse-square law.

The consequences were staggering. In the 1920s **Edwin Hubble** found Cepheids in the "spiral nebula" Andromeda, applied Leavitt's Law, and proved it was a separate galaxy far outside the Milky Way — ending the debate over whether the universe was one galaxy or many. He then used the same tool to show that more distant galaxies recede faster: the expanding universe. Every rung of the modern cosmic distance ladder still traces its ancestry to the pattern Leavitt spotted on photographic plates.

This project is that logic in miniature. CY Aqr is far too close to belong to another galaxy, and it is an SX Phe rather than a Cepheid — but the machinery is identical, and its class has its own period–luminosity relation. Period → luminosity → distance, verified against an independent yardstick. The same reasoning Hubble used to measure the universe, run on a backyard telescope.

The modern ladder, and where the pulsators sit

Today's distance ladder is a chain of methods, each calibrated by the rung below. At the base sit the **geometric** anchors, which use pure geometry and assume nothing about luminosity: **parallax** (the apparent shift of a star as Earth orbits the Sun — the method Gaia uses, and this project's answer key), plus water-**maser** distances to a few galaxies and **eclipsing-binary** distances. Above them, the **standard candles** whose luminosity is inferred: Cepheids, RR Lyrae, SX Phe, the tip of the red-giant branch, and at the summit Type Ia supernovae, which reach across the observable universe. A separate route — the sound horizon imprinted on the cosmic microwave background — predicts the expansion rate from early-universe physics. The two routes disagree slightly (the "Hubble tension"), a live problem at the frontier — and it turns out to hinge on whether the lowest rungs, the ones calibrating the candles, are correct. The same period–luminosity logic validated here, several rungs up, sits at the centre of the biggest open question in cosmology.

Part V · The Methods

The measurement pipeline in depth — how each number is actually computed.

V.0 The reference calibration: how Cohen & Sarajedini derived the relation

Everything in this project is measured against one published relation, so it is worth understanding exactly what went into it — both to trust it and to see honestly how an amateur re-derivation differs.

Cohen & Sarajedini (2012), *SX Phoenicis period–luminosity relations and the blue straggler connection*, calibrated the relation used throughout this document:

$$M_V = -1.640(\pm 0.110) + (-3.389\pm 0.090)\cdot \log_{10}(P_f)$$

Their derivation rested on four choices, each different from the approach taken here — and understanding the differences is more instructive than the agreement.

The sample: cluster and dwarf-galaxy stars, not field stars. They drew SX Phe pulsators from Galactic globular clusters and Local Group dwarf galaxies. This is the opposite of the strategy available to a small telescope. Cluster SX Phe are numerous — well over a hundred are catalogued across eighteen clusters — but they are faint and packed into crowded fields. Field SX Phe, the bright isolated ones this project uses, number only about fourteen. Their sample size came from going where the stars are; ours comes from going where the photometry is possible.

The photometry: Hubble, not backyard telescopes. Their magnitudes came from the ACS Galactic Globular Cluster Treasury Project — deep, uniform Hubble Space Telescope imaging of 65 clusters in the F606W and F814W filters. Space-based imaging resolves individual stars in crowded cluster cores that ground-based photometry blends together, which is precisely what makes the cluster sample usable at all. This project instead uses AAVSO archival photometry: ground-based, contributed by many observers, but on *bright, isolated* targets where crowding is a non-issue.

The distances: isochrone fitting, not parallax. This is the deepest difference, and it is chronological. Cohen & Sarajedini published in 2012; Gaia's first data release came in 2016, and its high-precision parallaxes later still. Per-star geometric distances simply did not exist for them. Instead they used **cluster** distances: every SX Phe in a given globular cluster inherits that cluster's distance, determined by fitting the cluster's main sequence to theoretical isochrones. To validate those distances they fitted the main sequences of 46 clusters against a carefully chosen set of nearby, unevolved **subdwarfs** whose own distances were known independently. So their distance scale is a chain: subdwarf distances → cluster main-sequence fits → cluster distances → SX Phe absolute magnitudes.

Mode identification: the double-mode trick. A subtle problem haunts any PL calibration for these stars. The relation applies to the *fundamental* radial mode, but many pulsators ring in an overtone instead, at a shorter period — and from a single light curve you often cannot tell which. Mis-identifying the mode puts a star badly off the relation. Their solution was elegant: build the calibration on **double-mode pulsators**, stars pulsating in two radial modes simultaneously. The *ratio* of the two periods is a sharp diagnostic — approximately 0.77–0.79 for the first-overtone-to-fundamental ratio — so observing both modes identifies each unambiguously. Ambiguity is removed by construction rather than assumed away.

How this project's approach differs — and where it is arguably cleaner. The comparison is genuinely two-sided. Their sample is far larger and photometrically superior (Hubble), chemically homogeneous (pure Population II SX Phe), and mode-unambiguous (double-mode ratios). Ours is small (eight stars), ground-based, and deliberately mixes SX Phe with Pop I δ Scuti because the bright field population is not large enough to be choosy. But on one axis the amateur approach is more direct: **we have Gaia, and they did not.** Each of our stars carries an individual geometric parallax distance, with no dependence on isochrone models, subdwarf calibrations, or the assumption that a cluster's stars share one distance. Their distance scale is a multi-step inference; ours is one measurement of an angle. That the two approaches agree — a quality-weighted fit to eight field stars reproducing a Hubble-based cluster calibration within uncertainties — is a meaningful cross-check precisely *because* the methods share almost no assumptions.

One further note on scope. Their title pairs the PL relations with "the blue straggler connection" for a reason: because most SX Phe are blue stragglers (Part II), a calibrated PL relation for them doubles as a tool for studying stellar mergers and mass transfer in dense clusters. The relation this project uses to measure one star's distance was built, in part, to investigate how ancient stars collide and are reborn.

V.1 Recovering the period: the Lomb–Scargle algorithm

The problem. We have a list of brightness measurements, each with a timestamp, and we want the pulsation period. If the data were evenly spaced, a Fourier transform would do it. But astronomical data is *never* evenly spaced — clouds interrupt, dawn ends the night, weeks pass between sessions. The timestamps are a ragged, gap-riddled set. The **Lomb–Scargle periodogram** is the tool built for exactly this: it generalises Fourier analysis to unevenly sampled data.

How it works, conceptually. For each trial period, Lomb–Scargle asks: "if I fold all the data onto this period and try to fit a sinusoid, how well does it fit?" More precisely, it fits a sine-plus-cosine model at each trial frequency and measures how much of the data's variance that frequency explains, reporting a "power." A period that matches the star's true rhythm lines the data up into a clean wave — high power. A wrong period scatters the data — low power. The periodogram is the plot of power versus trial period; the tallest peak is the best-fit period.

Mathematically, at angular frequency ω the power is (in the standard normalisation) computed from the data (t_i, y_i) as:

$$P(\omega) = \tfrac{1}{2} \left\{ [\Sigma\, y_i \cos\omega(t_i-\tau)]^2 / \Sigma \cos^2\omega(t_i-\tau) + [\Sigma\, y_i \sin\omega(t_i-\tau)]^2 / \Sigma \sin^2\omega(t_i-\tau) \right\}$$

where τ is a time offset chosen at each frequency to make the estimator insensitive to the arbitrary zero-point of the clock (this is the trick that makes it robust to uneven sampling). The details are in the equations notebook; the point is that it is a well-defined recipe returning a power for every trial period, requiring no evenly-spaced data.

Aliasing — the central pitfall. The periodogram often shows *several* peaks, and the tallest is not guaranteed correct. Two causes: (1) **daily aliases** — observing only at night imposes a one-cycle-per-day sampling rhythm that beats against the true frequency, producing false peaks at predictable offsets; (2) **harmonics** — a non-sinusoidal sawtooth has power at the true period and at fractions of it, notably half. This project hit aliasing directly: a thin dataset (42 points over 3 days) gave a forest of near-equal peaks and a period 68–140% wrong, while a single 3-hour night, spanning under one full cycle, could not pin a period at all. The cure was more data: a 16-year archive with hundreds of dense nights whose alias patterns do not align, so the true peak reinforced and the aliases washed out — recovering the period to 0.6%. The defence is always to *plot the periodogram and look*, never to trust the maximum blindly.

Folding needs a period; recovering one needs cycles. A subtle but vital distinction the project made concrete. To *fold* data (wrap it onto one cycle) you only need a known period — it works even on less than one cycle of coverage, because folding is just arithmetic on the phase. To *recover* a period you need many elapsed cycles, so the algorithm can see the rhythm repeat and time its spacing. This is why a short single night gives a beautiful folded light curve but a poor recovered period — and why the observing plan calls for at least three full cycles (four-plus hours for CY Aqr).

V.2 The intensity mean: computing the right average brightness

The "mean apparent magnitude" fed into the distance calculation is **not** a simple average of the measured magnitudes. Getting it right requires two corrections, both rooted in physics.

Correction 1: average in flux, not magnitude. Magnitude is a logarithm of flux, so averaging magnitudes averages logarithms — giving a geometric mean of flux, not the arithmetic mean we want. The physically meaningful average brightness is the mean *flux*, converted back to a magnitude at the end. Each magnitude is turned into relative flux, the fluxes are averaged, and the result is converted back:

$$\langle V\rangle = -2.5 \cdot \log_{10}\left(\frac{1}{N} \Sigma\, 10^{-0.4\, V_i} \right)$$

Because flux is convex in magnitude, this flux-average always comes out *brighter* than the magnitude-average — a real, systematic effect of a few hundredths of a magnitude for a star swinging 0.7 mag.

Correction 2: weight by phase, not by sample count. The sawtooth light curve means the star spends more time near minimum (the slow decline) than near maximum (the fast rise). If you simply average all your frames, you oversample the faint phases where the star lingers, biasing the mean faint. The fix is to **fold on the period and bin uniformly in phase**: divide the cycle into equal phase bins, average the flux within each bin, then average the bins — so every part of the cycle gets equal weight regardless of how many frames happened to land there. The full recipe: fold → bin in phase → flux-average each bin → average the bins → convert to magnitude.

Coverage is the honest limit. No averaging scheme can recover a phase you never observed. If a night misses maximum light, the mean is biased by exactly the missing bright phases — the reason the pipeline reports phase coverage and flags empty bins rather than silently interpolating. Empty bins are *skipped*, never invented: the mathematics must not fabricate data for a phase that was not measured. In practice, folding the full multi-year archive gave 100% coverage and shrank the flux-vs-naive bias to a few thousandths of a magnitude — the correction is small precisely because the coverage is complete.

V.3 From brightness to distance: the distance modulus

The link between apparent magnitude m (measured) and absolute magnitude M (from the period–luminosity relation) is the **distance modulus**, derived from the inverse-square law. Flux falls as $F = L / 4\pi d^2$, and magnitudes are −2.5 log of flux ratios; comparing the star at its true distance d against its definition at 10 parsecs (where m = M) gives:

$$m - M = 5 \log_{10}(d) - 5 \quad \text{(pure geometry)}$$

Interstellar **dust** dims the star further, adding A_V magnitudes of extinction, an additive term. Solving for distance:

$$m - M = 5 \log_{10}(d) - 5 + A_V \;\rightarrow\; d = 10^{(m - M - A_V + 5)/5}$$

The extinction degeneracy. Dust that dims the star mimics extra distance, so any error in A_V maps straight into a distance error. For CY Aqr the sightline is clean (galactic latitude −47°, far from the dusty plane), so A_V is small — but "small and uncertain" still matters, because it is an additive offset to the modulus.

Which way does a brightness error push the distance? The intuition worth internalising: at a fixed *measured* apparent magnitude, assuming the star is intrinsically *brighter* forces it *farther away* — because a more luminous source has to be more distant for its light to spread out (inverse-square) to the modest brightness we actually see. This is exactly why CY Aqr's period–luminosity distance (about 448 pc) came out *larger* than Gaia's (413 pc): the relation predicts CY Aqr should be brighter than it truly is (it is a faint outlier), so the method over-estimates its distance. The gap is not measurement error — it is the star's genuine faintness misread as distance.

V.4 Uncertainties: propagation, and two kinds of error

Every uncertainty combines by the standard rule of **propagation in quadrature**: for a quantity f depending on independent inputs x, y, …, the variance is the sum of squared partial derivatives times each input's variance,

$$\sigma_f^2 = (\partial f/\partial x)^2\sigma_x^2 + (\partial f/\partial y)^2\sigma_y^2 + \dots$$

which is why independent errors add in squares rather than linearly. Applied to the distance modulus (a simple sum) and the exponential distance, this carries the magnitude errors into a fractional distance error.

The crucial distinction: two different uncertainties. The period–luminosity relation carries *two* conceptually separate errors that answer different questions:

- **Calibration uncertainty** (about 0.15 mag) — how well we know the mean line itself, from the errors on its fitted slope and intercept. This is the error on the *average* star's luminosity.

- **Intrinsic scatter** (about 0.20 mag) — the real spread of individual stars *around* that mean line, from star-to-star differences in mass, exact temperature, and metallicity that the relation cannot capture.

Which to use, and why it is the precision floor. To predict the *average* star you need only the calibration error. To predict *one specific star's* distance — which is what we always actually do — you need both added in quadrature (about 0.25 mag total), and the intrinsic scatter dominates. This is the astrophysical floor of the whole method: no amount of better photometry on a single star beats it, because the star genuinely sits where it sits relative to the mean relation. The only way to improve is to average *many* stars, where the scatter beats down as $1/\sqrt{N}$ — which is exactly what deriving one's own relation from a sample does.

V.5 Deriving the relation: weighted fitting and quality discounting

The project's culminating step re-derives the period–luminosity relation from scratch, using eight stars with real intensity means (from AAVSO) and independent absolute magnitudes (from Gaia parallaxes). The relation $M_V = a + b\cdot\log P$ is a straight line in ($\log P$, M_V) space, so the task is a **linear least-squares fit** — but a *weighted* one, and the weighting is where the care lies.

Why weight at all. The stars are not equally trustworthy. Each has a different absolute-magnitude uncertainty (dominated by its parallax precision), and some are better-behaved than others. An unweighted fit would let a noisy, distant, or contaminated star pull the line as hard as a pristine one, discarding real information. Weighted least squares gives each star a weight $w = 1/\sigma^2$ — the inverse of its variance — so tightly-known stars pull harder.

The combined uncertainty per star. Each star's σ combines its parallax-propagated magnitude error with the intrinsic scatter of the relation (V.4), added in quadrature. No star is allowed to pull with unrealistic force, because the 0.2-mag intrinsic floor is always present in its weight.

Quality discounting — the defensible way to down-weight. Beyond the statistical weight, the project assigns each star a quality grade (A/B/C) from objective metrics (photometric coverage, number of points, parallax precision, whether the entered period matches what the data recovers) plus two flags set in advance on physical grounds: pulsation *mode* (single vs. double, since double-mode stars risk period confusion) and *population* (SX Phe vs. Pop I δ Scuti, since the two sit on slightly offset relations). Lower-grade stars have their errors inflated — effectively down-weighting them — so the clean, single-mode, metal-poor stars dominate the fit.

The honest reasoning behind the discount, and its trap. Down-weighting by *pre-registered* quality criteria is legitimate: the grades were defined from physical reasoning before the fit was run, so they do not cherry-pick a desired answer. The trap to avoid is choosing a weighting scheme *because* it produces agreement with the expected result — a subtle confirmation bias. The test is: would you make the same analysis choice if the published answer did not exist? Quality weighting passes; omitting a single star by name because it sits far from the line does not. The most defensible presentation reports the *range* of results — unweighted, quality-weighted, and population-split — and treats the spread as the honest measure of systematic uncertainty. In this project the unweighted fit gave a slope of −2.2 and the quality-weighted fit −3.2, both consistent with the published −3.39 within the (large) uncertainties — and the difference between them *is* the result: it demonstrates how sample purity drives the answer, which is precisely why professional calibrations use clean, single-population samples.

Avoiding circularity. When testing the relation on a specific star (say CY Aqr), that star must be *held out* of the fit — otherwise it helps define the very line used to judge it, and any agreement is partly guaranteed. The clean procedure is to fit the relation from the *other* stars, then drop the target on as an independent test point. This is the same principle as a train/test split in statistics: never evaluate on the data used to fit.

Part VI · Reference Material

Key numbers, coefficients, and the sample — for quick lookup.

The period–luminosity relation

Cohen & Sarajedini (2012), calibrated on Population II SX Phe in globular clusters and dwarf galaxies:

$$M_V = -1.640(\pm 0.110) + (-3.389\pm 0.090)\cdot\log_{10}(P_{fundamental})$$

with P in days. Intrinsic scatter about 0.20 mag. For a fundamental-mode period, log P is negative ($P < 1$ day), so the negative slope yields a positive M_V — these are intrinsically modest stars, $M_V \approx +1.5$ to $+3$, far fainter than the −3 to −6 of classical Cepheids.

CY Aquarii key numbers

Quantity	Value	Source
Period (fundamental)	0.06103845 d (87.9 min)	AAVSO / literature
Intensity-mean V	10.83 - 10.89	AAVSO fold (this project)
Absolute magnitude M_V	+2.48 (predicted) / +2.71 (Gaia)	PL relation / Gaia
Gaia parallax	2.385 +/- 0.020 mas	Gaia DR3
Gaia distance (Bailer-Jones geo)	413 [409, 416] pc	Bailer-Jones DR3
PL-derived distance	~448 +/- 53 pc	this project
Agreement	+0.66 sigma (within 1 sigma)	this project
Galactic latitude	-46.9 deg (low extinction)	-
Colour (Gaia BP-RP)	0.495 (blue-white)	Gaia DR3

The reference sample (8 stars)

Eight bright field pulsators with rich AAVSO V-band coverage and good Gaia parallaxes, spanning a factor of ~2.4 in period. Grades: A = single-mode, Population II SX Phe, rich data, tight parallax; B = demoted for Pop I δ Scuti membership or double-mode pulsation.

Star	P (days)	M_V	Population	Mode	Grade
CY Aqr	0.06104	2.71	SX Phe (II)	single	A
ZZ Mic	0.06718	1.81	dSct (I)	single	B
DY Peg	0.07293	2.24	SX Phe (II)	single	A
GP And	0.07868	1.94	dSct (I)	single	B
AE UMa	0.08602	1.76	SX Phe (II)	double	B
YZ Boo	0.10409	1.67	dSct (I)	single	B
XX Cyg	0.13486	1.45	SX Phe (II)	single	A
DY Her	0.14863	1.21	dSct (I)	single	B

Self-derived fit (all 8, quality-weighted): $M_V = -1.40(\pm 0.69) + (-3.17\pm 0.65)\cdot\log P$ — consistent with Cohen & Sarajedini within uncertainties. Unweighted: slope −2.2 (population mixing flattens it). The spread between the two is the honest systematic.

Observing CY Aqr yourself — quick reference

- **Exposure:** *~10 s (test first; keep target and comps unsaturated, below ~50% full well); shorter if using a focal reducer.*
- **Cadence:** *continuous, back-to-back; bin in software afterward.*
- **Duration:** *4–5 hours, centred on transit — captures 3+ of the 88-minute cycles (enough for independent period recovery).*
- **From Kona:** *CY Aqr culminates near overhead (Dec +1.5°, latitude +19.6°) — long low-airmass window.*
- **Calibration:** *darks, flats essential at the 0.01–0.02 mag level.*
- **Colour:** *OSC green channel ≈ V; use same-colour comps or derive a transform. A single clean night gives a good mean even untransformed.*

Companion: equations_reference.ipynb carries every formula above with worked numerical examples and runnable code. This document is the narrative and conceptual reference; the notebook is the computational one.